

Modern Application Development Project Report

Placement Portal Application

1. Student Details

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About me: I am a Computer Science student interested in software development, data science, and building scalable web applications. I enjoy solving

algorithmic problems and am currently exploring the field of Agentic

AI.

2. Project Details

Problem Statement:

to build a Placement Portal Application web application that allows Admin (Institute), Company, and students to interact with the system based on their roles.

Approach:

The application was implemented as a server-rendered full-stack web application using

Flask as the backend framework. HTML5 , Jinja2 and BOOTSTRAP5 were used for the front-end.

3. AI/LLM Declaration

I used ChatGPT (OpenAI) to assist in this project. The extent of AI/LLM usage is approximately 25%, mainly for:

- Debugging
- Assisting with UI and styling using BOOTSRAP
- Troubleshooting flask configuration issues
- Code suggestions

The final implementation, logic, routing and implementations were done manually.

4. Technologies and Frameworks used

Flask	Core backend web framework
Flask-Login	User authentication and session management
Flask-SQLAlchemy	ORM for defining database models
Flask WTF	Form validation and CSRF protection
SQLite	Relational database
Jinja2	HTML template rendering on Server-Side
Bootstrap 5	UI components and responsive layout
Werkzeug	Password hashing and verification

5. Database Schema/ER diagram

Tables:

1. User: This table stores authentication and role-based access details for all users in the system.
2. StudentProfile Table: Stores additional information for users registered as students.
3. CompanyProfile Table: Stores details about companies participating in the placement process.
4. PlacementDrive Table: Represents job recruitment drives conducted by companies.
5. Application Table: Stores job applications submitted by students.

Relationships:

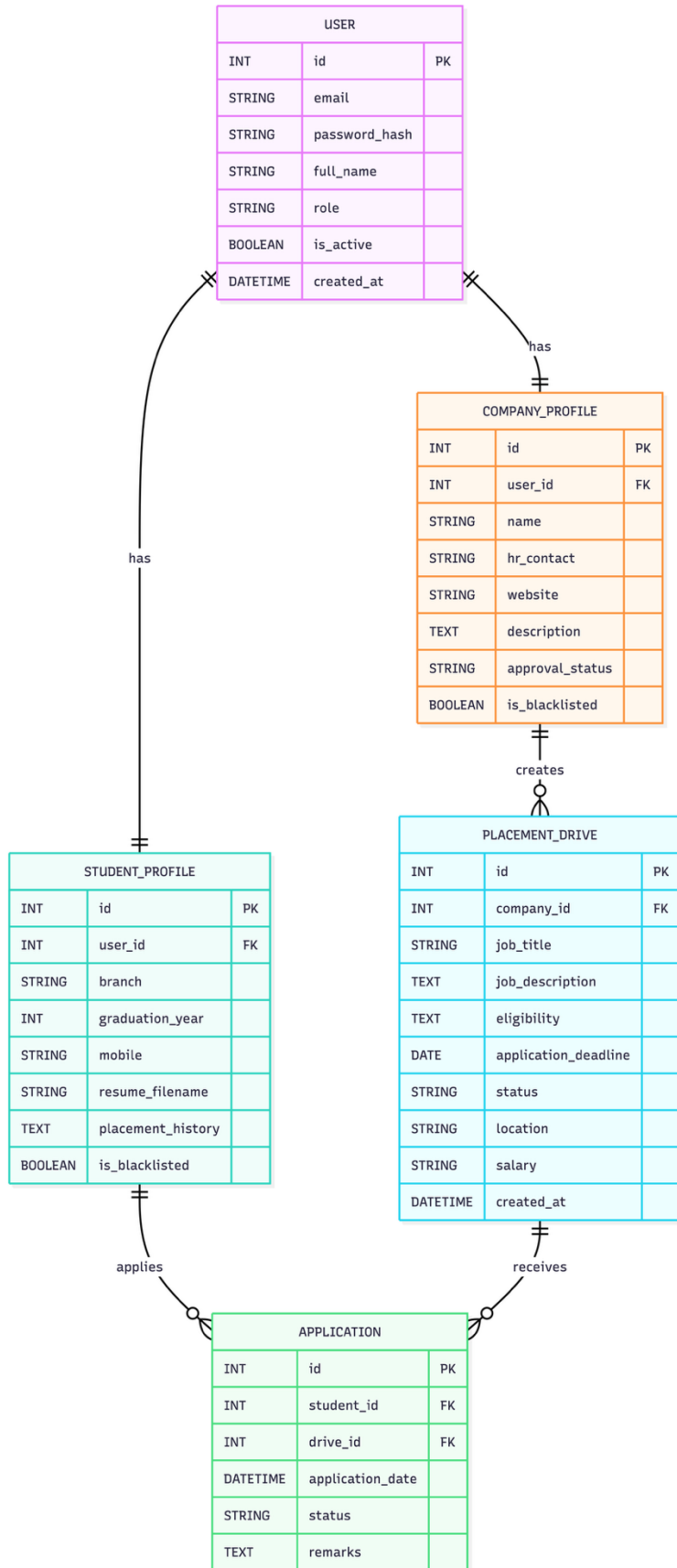
One User → One StudentProfile

One User → One CompanyProfile

One Student → Many Applications

One Company → Many Placement drives

One PlacementDrive → Many Applications



6. API Resource/Endpoints

Route	Method	Description
/	GET	Displays the home page with platform statistics and recent placement drives.
/login	GET, POST	Allows users (admin, company, student) to log in to the system.
/auth/login	GET, POST	Alternate route for user login.
/auth/logout	GET	Logs out the currently authenticated user.
/student/register	GET, POST	Allows a new student to register and create an account.
/company/register	GET, POST	Allows a company to register on the platform.

Student Endpoints

Route	Method	Description
/student/dashboard	GET	Displays student dashboard with available drives and application history.
/student/profile	GET, POST	View and update student profile details.
/student/drives/<drive_id>/apply	POST	Allows a student to apply for a placement drive.

Company Endpoints

Route	Method	Description
/company/dashboard	GET	Displays company dashboard with drives created and applicant statistics.
/company/drives/create	GET, POST	Allows a company to create a new placement drive.

Route	Method	Description
/company/drives/<drive_id>/edit	GET, POST	Allows a company to edit an existing placement drive.
/company/drives/<drive_id>/close	POST	Allows a company to close a placement drive.
/company/drives/<drive_id>/applications	GET	Displays all applications submitted for a specific drive.
/company/applications/<application_id>/status	POST	Updates the status of a student's application.

7. Video Presentation

Video link :

https://drive.google.com/drive/u/4/folders/196tRZWkt_YkayIfKrDITh4tsXX_NT_s